Data-integrated simulations of solute transport in the rodent brain using FEniCSx

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The glymphatic theory describes the extra-vascular transport and clearance of solutes in the brain. This solute transport is mediated by a set of barriers which are only partially understood. In this work, we emphasize the role of the pia membrane, the interface between the brain parenchyma and the cerebrospinal fluid (CSF), in determining solute transport in brain tissue. We use finite element simulations using FEniCSx with previously published data of tracer injection in 8 rats recorded by Single-Photon Emission Computed Tomography (SPECT) (Sigurdsson et al 2022). Parameter sweeps are performed using a quasi-Monte Carlo approach to estimate transport properties of this tracer, validating model outcomes with data recorded experimentally. Simulations use experimental data on the boundary of the domain, and we explore how the formulation of the boundary condition changes the transport properties predicted by the model. We show how FEniCSx can be combined with experimental data to provide insight into biological models and potentially inform new experiments.

References

Sigurdsson, et al., A spect-based method for dynamic imaging of the glymphatic system in

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